ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** three	Solid, liquid and gas.
2	**(c)** temperature is higher	Heat helps.
3	**(b)** collecting and storing rainwater	Collect rain for use.
4	**(c)** water vapour	Vapour is invisible water gas.
5	**(b)** the surface area is larger	More surface is exposed.
6	**(c)** closing taps when not in use	Prevent wastage.
7	**(d)** evaporation	Evaporation occurs at all temperatures.
8	**(b)** wind carries away water vapour	Wind speeds evaporation.
9	**(c)** flood	Flooding.
10	**(b)** condensation	Cooling of vapour.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Solid: ice (1); Liquid: water; gas: water vapour (1).	1+1
12	Continuous movement of water on the Earth (1); Evaporation, condensation and precipitation (1).	1+1
13	Heat speeds evaporation (1); Wind carries vapour away (1).	1+1
14	Change of liquid to vapour at any temperature (1); Occurs from the surface of liquid (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sun heats water bodies (1); Vapour rises and cools (1); It condenses into tiny droplets forming clouds (1).	1+1+1
16	Ice melts to water at 0 °C (1); Water heats up to 100 °C (1); It boils to steam (1).	1+1+1
17	Cooling of air (1); Condensation of vapour (1); Droplets near ground (1).	1+1+1
	OR: Evaporation from water bodies (1); Condensation into clouds (1); Precipitation as rain or snow (1).	1+1+1
18	Water evaporates (1); It goes into the air (1); Faster in heat (1).	1+1+1
	OR: Heat increases evaporation (1); Water vapour goes into the air (1); Level falls (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sun heats sea (1); Evaporation (1); Condensation to clouds (1); Precipitation (1); Water flows back to rivers and seas (1).	1+1+1+1+1
	OR: Drinking, cooking (1); Farming (1); Close taps (1); Fix leaks (1); Harvest rain (1).	1+1+1+1+1
20	Solid: ice (1); Liquid: water (1); Gas: vapour (1); Melting, freezing (1); Evaporation, condensation (1).	1+1+1+1+1
	OR: Rise in temperature (1); Boiling at 100 °C (1); Steam is formed (1); Cooling gives droplets (1); Further cooling gives ice (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing evaporation with **boiling**: evaporation happens at any temperature from the surface, boiling at about 100 degrees C.
- Mixing condensation (vapour to liquid) with evaporation (liquid to vapour).
- Calling the white cloud over a boiling kettle water vapour; vapour is invisible, the white cloud is tiny droplets.
- Saying that drops on a cold glass come from inside the glass; they come from **vapour in the air**.
- Writing that humid air helps drying; more moisture in the air makes evaporation slower.
- Forgetting that ice melts and water freezes at the same temperature, 0 degrees C.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** air already has much water vapour	High humidity.
2	**(c)** drought	Drought.
3	**(a)** freezing	At 0 °C.
4	**(d)** humidity	Humidity.
5	**(c)** ice is lighter than water	It is less dense.
6	**(b)** melting	At 0 °C.
7	**(d)** the Sun's heat	The Sun drives evaporation.
8	**(b)** 0 °C	Melting point.
9	**(d)** 100 °C	At normal pressure.
10	**(a)** precipitation	Falling water.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water stored beneath the Earth's surface (1); Taken by wells and hand pumps (1).	1+1
12	0 °C (1); Ice changes to water (1).	1+1
13	Vapour changes to liquid on cooling (1); Droplets on a cold glass (1).	1+1
14	Collecting rainwater and storing it (1); Used to recharge groundwater (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heat from the Sun (1); Wind removes vapour (1); Evaporation increases (1).	1+1+1
16	Rain falls on roofs (1); Pipes lead it to a tank or pit (1); It is stored or soaks into the ground (1).	1+1+1
17	May contain germs (1); Has dirt (1); May cause diseases (1).	1+1+1
	OR: Air has water vapour (1); Vapour touches cold glass (1); It condenses (1).	1+1+1
18	Evaporation occurs at any temperature (1); Boiling occurs at a fixed temperature (1); Evaporation is at the surface, boiling throughout (1).	1+1+1
	OR: Rivers (1); Lakes (1); Groundwater (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Spread wet cloths (1); One in sun, one in shade (1); Sunny one dries first (1); Larger area dries faster (1); Wind increases speed (1).	1+1+1+1+1
	OR: Water vapour in air (1); High humidity slows drying (1); Low humidity speeds drying (1); Temperature matters (1); Wind also matters (1).	1+1+1+1+1
20	Take a glass of cold water (1); Keep in the room (1); Droplets appear outside (1); Vapour in air condensed (1); Not leakage (1).	1+1+1+1+1
	OR: Roof collects water (1); Pipes guide it (1); Stored in a tank (1); Recharges groundwater (1); Saves water (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing evaporation with **boiling**: evaporation happens at any temperature from the surface, boiling at about 100 degrees C.
- Mixing condensation (vapour to liquid) with evaporation (liquid to vapour).
- Calling the white cloud over a boiling kettle water vapour; vapour is invisible, the white cloud is tiny droplets.
- Saying that drops on a cold glass come from inside the glass; they come from **vapour in the air**.
- Writing that humid air helps drying; more moisture in the air makes evaporation slower.
- Forgetting that ice melts and water freezes at the same temperature, 0 degrees C.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** early morning on cool surfaces	Vapour condenses.
2	**(c)** tiny water droplets and ice crystals	Condensed vapour.
3	**(d)** groundwater	Stored below the surface.
4	**(a)** water vapour condenses near the ground	Tiny droplets near the surface.
5	**(d)** water cycle	Hydrological cycle.
6	**(b)** water table	The water table.
7	**(b)** river	Rivers carry fresh water.
8	**(c)** condensation of water vapour from air	Vapour cools on the glass.
9	**(b)** groundwater	Wells reach the water table.
10	**(c)** salty	It contains dissolved salts.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water falling from clouds (1); Rain, snow, hail (1).	1+1
12	Temperature (1); Wind or surface area (1).	1+1
13	Ice is less dense (1); It occupies more volume than the same mass of water (1).	1+1
14	Fresh water is limited (1); Needed for all living things (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Upper level of groundwater (1); Wells and pumps tap it (1); It supports water supply (1).	1+1+1
16	Ice is lighter (1); It forms a layer on top (1); Water below stays liquid for life (1).	1+1+1
17	Rivers carry salts from rocks (1); Evaporation leaves salt behind (1); Salts accumulate (1).	1+1+1
	OR: Close taps when not in use (1); Repair leaks (1); Reuse water for plants (1).	1+1+1
18	Flood: too much water (1); Drought: lack of rain (1); Management: dams, harvesting, afforestation (1).	1+1+1
	OR: Heating: ice to water to vapour (1); Cooling: vapour to water to ice (1); Temperature changes cause this (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Used for drinking and farming (1); Wells, hand pumps (1); Rainwater harvesting (1); Trees help soak water (1); Avoid overuse (1).	1+1+1+1+1
	OR: Needed for life (1); Used in farming (1); Industry (1); Many substances dissolve (1); Conserve it (1).	1+1+1+1+1
20	Heavy rain causes floods (1); Floods damage homes (1); Little rain causes drought (1); Crops fail (1); Suggest remedies (1).	1+1+1+1+1
	OR: Rivers and lakes (1); Sea is salty (1); Fresh water used for drinking (1); Sea water for salt (1); Fresh water is limited (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing evaporation with **boiling**: evaporation happens at any temperature from the surface, boiling at about 100 degrees C.
- Mixing condensation (vapour to liquid) with evaporation (liquid to vapour).
- Calling the white cloud over a boiling kettle water vapour; vapour is invisible, the white cloud is tiny droplets.
- Saying that drops on a cold glass come from inside the glass; they come from **vapour in the air**.
- Writing that humid air helps drying; more moisture in the air makes evaporation slower.
- Forgetting that ice melts and water freezes at the same temperature, 0 degrees C.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.